

IMPLEMENTATION GUIDE

StockPilot *setup guide*

Autonomous inventory and replenishment agent

This is the system from LinkedIn. One engine forecasts demand for every product, computes a reorder plan from real inventory math, posts it to Slack for a person to approve, then writes the approved order back to your store. This guide gets the free template running on your own n8n.

What you are getting

One engine, two modes

A demo mode that runs on a CSV with no accounts, and a connected mode that runs live on Shopify. Same brain.

Real forecasting

An exponential smoothing baseline plus gradient boosting, with a walk forward backtest that picks the best model per SKU and reports MAE and MAPE.

Inventory math, not guesses

Safety stock, reorder point, and economic order quantity decide when and how much. The model never invents a number.

Risk before it happens

Days of cover, stockout probability, and an anomaly flag surface what runs out before a reorder can land.

Human approval

The plan arrives in Slack with approve and reject buttons. Nothing is ordered without a click.

Live dashboard

A calm operations view: the weekly plan, forecasts, the four status inventory map, ABC concentration, and the run history.

What you need before you start

All of these have free tiers that are enough to run and test the system.

n8n

Shopify

NVIDIA NIM

Slack

Python 3.11+

n8n Self hosted or cloud. Orchestrates the schedule, the Shopify pull, the engine call, the Slack approval, and the write back.

Shopify A store, where a free development store works. Provides products, live inventory, and order history, and receives the approved draft order.

NVIDIA NIM The free model that ranks the plan and writes the summary. Get a key at build.nvidia.com.

Slack Receives the plan and sends the approve and reject click back to n8n.

Python Runs the engine locally, the forecasting and the inventory math, at no cost.

Setup, step by step

STEP 1

Run the engine

```
python -m venv .venv && source .venv/bin/activate
pip install -r requirements.txt
cp .env.example .env          # add your NIM_API_KEY
uvicorn api.main:app --host 0.0.0.0 --port 8000
```

Open localhost:8000/health and you should see status ok with NIM configured. Try the whole demo with no accounts at all using `python scripts/run_demo.py`.

STEP 2

Make the engine reachable from n8n

n8n in the cloud, or on another machine, cannot see your localhost. Open a tunnel:

```
cloudflared tunnel --url http://localhost:8000    # or: ngrok http 8000
```

Copy the public https URL. For a permanent setup, deploy the repo to Render, Railway, or Fly and run the same uvicorn command.

STEP 3

Import the workflow

In n8n, choose Import from File, then `n8n/stockpilot_workflow.json`. It imports deactivated on purpose. Paste your public engine URL into the two Engine nodes, `/run` and `/plan`.

STEP 4

Connect credentials

Add each one once, then assign it on the nodes that show a warning.

Shopify. Use the Shopify Access Token API credential, or OAuth2. Scopes: `read_products`, `read_orders`, `read_all_orders`, `read_inventory`, `write_draft_orders`. Set the node Authentication to match the credential type.

Slack. Use the Slack API credential with your bot token `xoxb-...` and the scope `chat:write`. Invite the bot to your channel. Set the channel in Build Slack Message to the channel ID, the value like `C...`, not the name.

NVIDIA NIM. Set `NIM_API_KEY` in the engine `.env`, not in n8n. The engine calls it once per run.

STEP 5

Wire the Slack buttons

Save, then Activate the workflow. Copy the Slack Interactivity Webhook node Production URL into your Slack app, under Interactivity and Shortcuts, as the Request URL.

STEP 6

Test the loop

Click Execute Workflow. A ranked plan posts to Slack. Click Approve and the engine plan is fetched, a draft order tagged **stockpilot** is created in Shopify, and a confirmation posts back to Slack.

STEP 7

Run the dashboard, optional

The dashboard is optional and renders on its own sample data.

```
cd dashboard && npm install && npm run dev # http://localhost:5173
```

To go live, set **VITE_USE_SAMPLE=false** and point **VITE_API_BASE** at your engine in **dashboard/.env**.

COMMON GOTCHAS

1. **Empty store.** Forecasting needs months of order history, so a fresh development store gives a thin plan. Seed it with historical orders to see real numbers.
2. **Slack channel.** Post by channel ID, not by name, since name resolution needs an extra read scope. Invite the bot to the channel or the messages bounce.
3. **NIM rate limits.** The engine sends the whole plan in one call rather than one per SKU, and falls back to a second provider when you set the fallback values in the env file.
4. **The model touching numbers.** Every quantity it returns is checked against the source and repaired if it drifted. The model only ranks and writes.
5. **Shopify cost and lead time.** Shopify's product payload has neither, so the workflow defaults cost to 40 percent of price and lead time to 7 days. Override per SKU through metafields or a CSV for exact orders.
6. **Tunnel URL changes.** A free tunnel gives a new URL on restart, so update the two Engine nodes, or deploy for a stable URL.

WANT IT DONE FOR YOU

We build, configure, and deploy the whole system

The template is the engine. Standing it up for a real store takes configuration, tuning, and hardening. NeuragenceAI handles all of it, fully managed and ready for production.

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|--|--|
| + A custom branded dashboard | + Full local or cloud deployment |
| + Your store connected and tested | + Suppliers, costs, and lead times tuned |
| + Forecast models fitted to your history | + Write back to your store configured |
| + The Slack approval flow wired end to end | + Production hardening and ongoing support |

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